

Summary of Revisions Following Peer Review and Post-Review Audit

PhysioRisk: an interpretable survival-modeling framework for separating demographic and physiological mortality risk

Gregory S. Allen | Research Square preprint rs-9829732 | Version 2

Context. Version 2 was prepared following external peer review of the journal submission and a subsequent reconstruction and audit of the analytical workflow. Peer review questioned the mathematical formulation of the composite TotalRisk score, the chronology of model development, uncertainty in the reported performance estimates, cohort flow and missing-data handling, comparator definition, calibration and proportional-hazards assessment, reproducibility, and the distinction between internal temporal and external validation. Addressing these concerns led to additional analyses and to a broader audit of the preserved computational record.

The post-review audit also identified an additional technical issue not fully recognized in Version 1: the Cox evaluation underlying the originally reported PhysioRisk hazard ratio had not converged. Version 2 replaces that estimate with a convergence-controlled evaluation. The audit also established more precisely what can, and cannot, be demonstrated from the surviving development record. The revisions below therefore include both direct responses to peer-review concerns and corrections uncovered while carrying out that review-driven audit.

Revisions addressing peer-review concerns

1. Composite risk score reconstructed.

Peer review correctly identified that the Version 1 unit-weight TotalRisk, formed by adding DemoRisk to standardized PhysioRisk, was not generally equivalent to a jointly estimated Cox linear predictor. Version 2 therefore replaces it as the primary composite with $\text{CalibratedTotalRisk} = 1.0586578128 \times \text{DemoRisk} + 0.3563514639 \times \text{PhysioRisk}$. Both coefficients were estimated jointly using only the 1999–2010 development cohort and then fixed before 2011–2016 temporal validation. The historical unit-weight TotalRisk is retained only for provenance. In temporal validation, CalibratedTotalRisk had C-index 0.843 (95% CI 0.822–0.863) and exceeded DemoRisk by $\Delta C = 0.01628$ (95% CI 0.01109–0.02199).

2. Development chronology and provenance clarified.

Peer review also noted that the original manuscript did not make the model-development sequence sufficiently auditable. The preserved computational record is sufficient to reproduce the final scoring and evaluation path, but not to establish the complete chronology of every earlier decision leading to selection of the final two physiological axes. Version 2 states this limitation explicitly rather than asserting that all historical model-selection decisions were entirely development-only. This limitation is distinguished from the revised CalibratedTotalRisk recombination coefficients, which were demonstrably estimated in development and frozen before temporal evaluation.

3. Uncertainty and validation diagnostics expanded.

Version 2 adds participant-level bootstrap confidence intervals for C-indices and paired intervals for prespecified model contrasts; 5- and 8-year calibration; observed-to-expected ratios; horizon-specific calibration slopes and intercepts; IPCW Brier scores; Uno C; cumulative/dynamic AUC; reverse Kaplan–Meier follow-up; numbers at risk; and proportional-hazards diagnostics. Mean predicted and observed mortality were closely aligned at 5 years (4.610% vs 4.653%) and 8 years (8.571% vs 8.826%). CalibratedTotalRisk nevertheless violated strict proportional hazards, with its association strengthening over follow-up, so a single constant hazard ratio is interpreted cautiously.

4. Cohort flow, missingness, comparator definition, and reproducibility strengthened.

The revised manuscript reconstructs the full cohort flow from 92,062 source-cycle participants to 53,255 mortality-eligible participants, 19,829 biomarker-complete participants, 13,090 development participants, and 6,739 temporal-validation participants. It reports the characteristics of excluded mortality-eligible participants and cycle-specific biomarker missingness, and makes clear that the analysis is an unweighted complete-case model evaluation rather than a population-representative NHANES estimate. PhenoAge NCP is explicitly described as a no-CRP proxy rather than the complete published PhenoAge model, and the main comparison is evaluated on the same participants with common missing-data sensitivity checks. Supplementary Tables S1–S9 provide the numerical diagnostics and provenance supporting the revised manuscript.

Additional correction identified during the post-review audit

5. PhysioRisk hazard-ratio estimate corrected.

During post-review reconstruction, the unpenalized Cox evaluation underlying the Version 1 PhysioRisk HR was found not to have converged. Version 2 replaces that estimate with a convergence-controlled Cox evaluation and reports HR 1.387 (95% CI 1.338–1.439) per temporal-validation-cohort standard deviation. PhysioRisk discrimination and age correlation remain C-index 0.654 (95% CI 0.624–0.685) and $r=0.106$, respectively. The correction materially changes the magnitude of the hazard gradient but does not change the conclusion that age- and sex-adjusted physiological variation retains mortality information.

Interpretation revised in light of review and audit

6. Claims narrowed and the conceptual contribution stated more precisely.

Version 2 no longer frames the main result as predictive superiority. CalibratedTotalRisk and PhenoAge NCP had no established difference in discrimination ($\Delta C=0.00205$; 95% CI -0.00326 to 0.00748). Likewise, the PhysioRisk-versus-PhenoAge NCP acceleration difference was suggestive rather than conclusive. The revised paper therefore emphasizes the interpretable separation of demographic mortality structure from age- and sex-adjusted physiological variation. PhysioRisk is presented as a statistical physiological-risk component, not as a definitive biological-age measure or causal estimate of aging.

7. Validation status and limitations made explicit.

The 2011–2016 evaluation remains internal temporal validation within NHANES, not independent external validation. Version 2 states this consistently and avoids claims of transportability or clinical utility. It also expands the limitations concerning complete-case selection, historical model-selection provenance, the no-CRP comparator, fixed-horizon calibration uncertainty, proportional-hazards violation, absence of subgroup and decision-utility analyses, and the smaller risk-set tail at 8 years.

Overall effect of the revision

The review-driven reconstruction materially strengthened the manuscript and narrowed several claims. The central finding remains: age- and sex-adjusted physiological variation retains mortality information and adds incremental discrimination when recombined with demographic risk. The revised analysis does not establish superiority over PhenoAge NCP, and independent external validation remains the principal unresolved question. Version 2 is intended to make the analytical record, corrections, uncertainty, and remaining limitations substantially more transparent than in Version 1.

This is an author-prepared summary of changes made after external peer review and subsequent audit; it is not a reproduction of the confidential reviewer reports.